Targeted Regulation and Noncompliance: Evidence from Interest Rate Caps*

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November 2, 2026

Abstract

Targeted regulation relies on classification, creating a selective noncompliance margin that can undermine policy objectives. We investigate how noncompliance reshapes the effects of targeted regulation in the context of interest rate caps, which trade off credit access and borrower protection. Using a reform that tightened caps on agricultural loans in Ecuador, we document strategies used to bypass the regulation and charge interest rates above the cap. Noncompliance disproportionately affects borrowers with a single lending relationship. We develop an empirical model to quantify the distributional effects, finding that noncompliance mitigates the loss in credit access induced by the cap, particularly for new borrowers.

Keywords: regulation, compliance, interest rate caps, agricultural credit
JEL Codes: G21, G28, L13, L51, Q14

*Preliminary draft.

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1 Introduction

Regulation is often targeted toward specific agents, products, or activities for efficiency, equity, or administrative feasibility reasons. For instance, tax rates vary by products, rent controls are often defined by location or property type, and labor regulation applies depending on whether a worker is a firm's employee or an independent contractor. Because fully personalized regulation is infeasible, implementation requires classification. When agents can influence classification, targeted regulation creates a selective noncompliance margin that can undermine policy objectives and generate unintended consequences.

How does noncompliance reshape the effects of targeted regulation? We study this question in the context of interest rate caps, a policy tool widely used around the world and often targeted to specific sectors or loan types (Maimbo and Henriquez Gallegos, 2014; Ferrari et al., 2018). Caps involve a trade-off between borrower protection and credit access. On the one hand, caps constrain the ability of lenders to exercise market power by setting high prices. On the other hand, by lowering prices, caps reduce lending profitability, which can induce lenders to ration credit. The effects of the regulation are heterogeneous across borrowers, since those who are able to borrow at lower prices benefit from it, while those who lose access are harmed. Noncompliance alters both sides of the trade-off. For borrowers who would otherwise obtain a loan under the cap, it removes the gains from lower prices. For those who would otherwise be rationed, it provides credit access. Because lenders choose which borrowers receive noncompliant loans, the distributional effects of the cap can be mitigated or exacerbated. In this paper, we quantify these effects.

We exploit a reform that introduced an interest rate cap on agricultural loans in Ecuador in 2019. With the intended purpose of increasing credit access for agricultural borrowers, the reform created a new regulatory microcredit subsegment for agricultural loans. These loans had a cap below the one that applied to non-agricultural loans. Since enforcement of the regulation depends on how lenders classify loans across regulatory subsegments, reporting loans that finance agricultural activities under non-agricultural subsegments creates scope for noncompliance.

We document this noncompliance using loan-level administrative data on cooperatives' originations, which allow us to track each borrower's loans between 2017 and 2020.[1] We define noncompliance as granting a loan to finance agriculture at an interest rate above the agricultural cap. We document two strategies used to evade the cap. We call the first, *subsegment misclassification* which consists of classifying a loan that finances an agricultural activity under a non-agricultural regulatory subsegment. Because cooperatives report both the loan's economic activity and its regulatory subsegment, we observe this practice directly in the data. For instance, we observe loans for maize cultivation classified under non-agricultural subsegments. We find that 34.9% of loans tied to agricultural activities are classified under non-agricultural subsegments and carry an interest rate above the agricultural cap.

The second, which we call *activity reclassification*, consists of reporting a loan that finances an agricultural activity as financing a non-agricultural activity, and classifying the loan under a non-agricultural subsegment. In this case, we observe both a non-agricultural activity (for instance, manufacturing) and a non-agricultural subsegment. Since the true activity that the loan finances is unobserved, we rely on borrowers who obtain multiple loans over the period of analysis. We say a borrower switches activities when a loan is associated with an agricultural activity and the previous one with a non-agricultural activity, or vice versa. We interpret pre-reform switching as reflecting genuine changes in activity, and attribute the post-reform increase to reclassification. We find that the overall rate of activity switching did not increase after the introduction of the agricultural cap, but that the share of switches out of agriculture rose from 57% to 68.9%, an increase of 21%. Moreover, borrowers who switch out of agriculture pay higher interest rates than those who switch into agriculture or do not switch. We estimate that 4% of the loans are reclassified as financing non-agricultural activities and carry interest rates above the agricultural cap.

We find evidence of noncompliance being primarily a lender-level decision. Lender identity explains more of the variation in both strategies than borrower or location charac-

[1] Due to data limitations, we do not study noncompliance in the banking sector.

teristics, and lenders that use one strategy also tend to use the other. Noncompliance is more common among lenders with higher pre-reform interest rates, which have the most to lose from complying with the cap, and less common among lenders with larger market shares. Competition is associated differently with each strategy: subsegment misclassification is more common in markets with more lenders, while activity reclassification is less common. These patterns suggest that noncompliance responds to market structure and is concentrated among single-relationship borrowers. This motivates our treatment of noncompliance as a strategic lender choice given borrower heterogeneity.

To understand the effects of noncompliance, we develop and estimate an empirical model in which cooperatives extend loan offers to borrowers, who then accept or decline the offer. On the demand side, borrowers consider the interest rate, cooperative characteristics such as the distance to the closest branch and the number of branches in their province, and the intensity of their relationship with the cooperative. On the supply side, cooperatives make a discrete choice between complying with the cap, avoiding the regulation through activity reclassification or subsegment misclassification, or not extending an offer. To do so, they weigh the expected profits of each contract against the costs of making an offer and of noncompliance.

We estimate the model distinguishing between repeat borrowers, those who obtained credit from a cooperative in the past, and new borrowers. We find that extending an offer to a new borrower is more costly, and that noncompliance is also more costly when the borrower is new. Noncompliance costs are lower in markets with more cooperatives and for lenders with larger market shares, on average.

Using the estimated model, we quantify the consequences of noncompliance by simulating full compliance under the agricultural cap. Welfare effects depend on the trade-off between lower interest rates and credit rationing. Under full compliance, borrowers who keep an offer pay lower rates, but lenders stop serving borrowers for whom lending at the cap is not profitable. For repeat borrowers, borrowing falls by 4% and their consumer surplus increases by about 1%. New borrowers face most of the rationing. Borrowing falls

by 25% and their consumer surplus decreases by 9%. These results imply that noncompliance mitigated credit rationing induced by the cap, particularly among new borrowers.

This paper contributes to two strands of literature. First, it relates to research on interest rate caps. The trade-off between credit access and borrower protection has been documented by Madeira (2019) and Cuesta and Sepúlveda (2021) who find that borrowers who still receive loans pay less but lenders ration out riskier borrowers. Cherry (2025) finds that caps raise average prices by driving out lenders with superior screening technology, and affects borrowers disproportionately. Furthermore, in markets with lender market power, caps can increase borrowing without rationing (Fekrazad, 2020; Kuroishi et al., 2025). The effects of caps also depend on local market concentration, which shapes how lenders reallocate credit across borrowers (Burga et al., 2025). These studies share two assumptions. Regulation applies uniformly within a market, and lenders comply. We relax both. In our setting, the cap targets one sector, and lenders choose whether to comply by choosing how to classify loans. Noncompliance changes the access-protection trade-off. It mitigates rationing, since lenders keep lending above the cap. It also weakens protection, and the burden falls unevenly across borrowers.

Second, the paper relates to research on how firms respond to regulation that depends on classification or thresholds. A broadly documented result is that firms adjust real behavior to avoid regulatory thresholds. Becker and Henderson (2000) and Garicano et al. (2016) find exactly this, as firms shift plant location, size, and employment to stay below thresholds that trigger stricter rules. Dafny (2005); Begley et al. (2017); Geruso and Layton (2020) find that firms change how they report instead of what they do, misreporting or reclassifying to lower regulatory costs or raise payments. Furthermore, evasion responds to enforcement intensity in tax and environmental settings (Kleven et al., 2011; Zou, 2021). Our setting features a similar margin, as enforcement costs differ across noncompliance strategies. We study a market where lenders manipulate classification directly. Our contribution is to show that this margin interacts with market power. Noncompliance is concentrated among lenders with high pre-reform rates and among borrowers with a single lending relationship. Targeted regulation therefore creates classification margins that

reshape who bears its incidence.

The closest paper to ours is Ontaneda (2026), which studies the same reform, focusing on how targeted interest rate caps affect lenders' loan offers and borrowers' acceptance in the targeted sector. While Ontaneda (2026) takes noncompliance as given, this paper models it as a strategic lender choice, which allows us to quantify how it reshapes the distributional effects of the cap.

The rest of the paper proceeds as follows. Section 2 describes the institutional context and the data. Section 3 defines noncompliance, measures the two strategies lenders use to evade the cap, and documents how noncompliance varies across lenders, markets, and borrowers. Section 4 develops an empirical model of demand and lender contract choice. Section 5 describes the estimation and presents the structural results. Section 6 evaluates the effects of noncompliance and section 7 concludes.

2 Institutional Context and Data

In this section we provide details about the credit market and the interest rate regulation in Ecuador. We also describe the data sources and present summary statistics.

2.1 The credit market and the interest rate regulation in Ecuador

Ecuador's financial sector is relatively underdeveloped. The share of adults who borrow from a formal financial institution is similar to the developing country average (Development Research Group, 2022). We focus on agricultural lending to small and medium firms, which the regulation classifies as microcredit (see below). This market is served by cooperatives, private banks and a public bank. In 2018, cooperatives provided 55% of agricultural microcredit funds. Private banks accounted for only 12%. A large retail public bank that offers subsidized loans accounted for roughly one third. Due to data

availability the analysis is restricted to cooperatives. As documented in Ontaneda (2026), lenders in this market actively reach out to borrowers with loan offers.

Interest rate regulation. Since 2007, Ecuador's credit market has operated under a system of segment-specific absolute interest rate caps. The regulation divides the market into segments, such as commercial, consumer, microcredit, and mortgage loans, and assigns a separate cap to each. Microcredits are loans for productive purposes extended to firms with annual sales of up to $100,000. They are further divided into three subsegments according to the borrower's outstanding debt: *minorista* (up to $1,000), *acumulación simple* (more than $1,000 and up to $10,000), and *acumulación ampliada* (more than $10,000). Each subsegment has its own cap. Since 2018, caps have also varied by the type of lender originating the loan, with banks and large cooperatives facing lower caps than small cooperatives.[2]

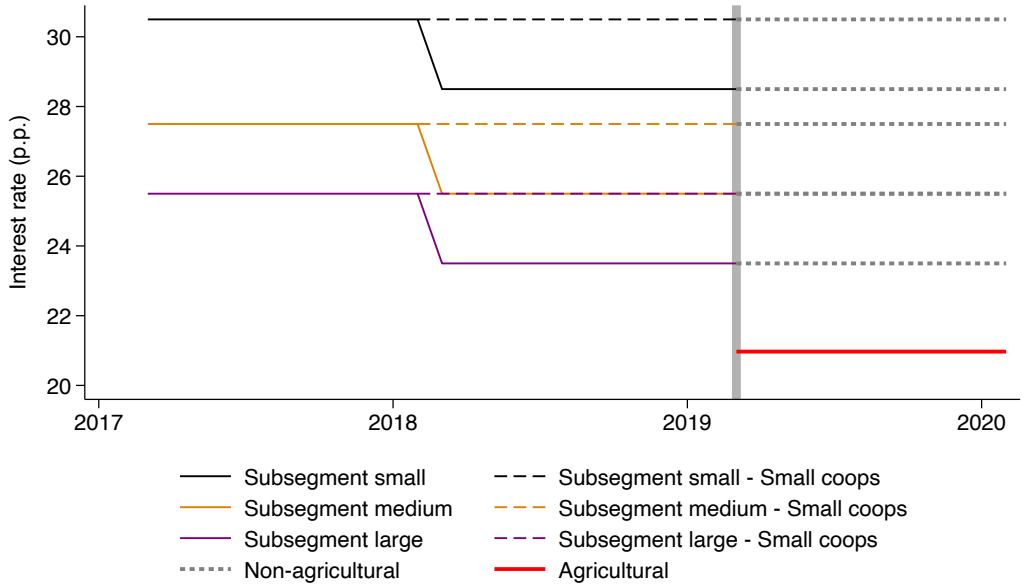
Figure 1 shows the evolution of microcredit interest rate caps across subsegments and lender types between 2017 and 2020. The reform we study, introduced in March 2019, created a separate cap for microcredits financing agricultural and livestock activities (hereafter, agricultural loans), with the stated goal of expanding credit access in the agricultural sector. The cap was set at 20.97%, between 2.5 and 9.5 percentage points below the cap of pre-existing microcredit subsegments, which continued to apply to non-agricultural loans. The originating lender reports each loan's segment, subsegment, and economic activity. Therefore, whether a loan falls under the agricultural cap depends on how the lender classifies it.

2.2 Data

Our primary data source is a loan-level administrative dataset covering cooperative originations from 2017 to 2020, obtained from the cooperatives' supervisory agency (Super-

[2]Cooperatives are classified into five tiers by asset size with tier 1 corresponding to large cooperatives.

Figure 1: Evolution of Microcredit Interest Rate caps



Note: This figure displays the evolution of microcredit interest rate caps, 2017–2020. The gray vertical bar marks the introduction of the agricultural cap in March 2019. Before the reform, three caps were defined based on borrowers' outstanding debt: (i) Subsegment small (*minorista* for borrowers with debt of up to \$1,000); (ii) Subsegment medium (*acumulación simple* for borrowers with debt of more than \$1,000 and up to \$10,000); and (iii) Subsegment large (*acumulación ampliada* for borrowers with debt of more than \$10,000). In 2018, the caps for banks and large cooperatives were reduced (solid lines), while they remained unchanged for smaller cooperatives (dashed lines). After the reform, the red line marks the agricultural subsegment cap, while gray dotted lines represent pre-existing subsegments that applied to non-agricultural loans.

intendencia de Economía Popular y Solidaria, SEPS).[3] For each loan, we observe the effective interest rate, the regulatory subsegment under which the loan was classified, the economic activity code associated with the loan, and a borrower identifier that links loans to the same borrower over time. Together, the interest rate, subsegment, and activity code are what allow us to measure noncompliance. We define agricultural loans as those whose activity code falls under ISIC code A01.

A second dataset contains monthly snapshots of cooperatives' outstanding loans over the sample period. It includes loans originated before 2017 that were still outstanding in 2017, which lets us extend each borrower's credit history before the start of the origination

[3]We also have data on loans originated by private banks and the public bank, but these data are not at the loan level and do not include a borrower identifier, which prevents us from studying noncompliance in the banking sector.

data. This longer history is critical for measuring noncompliance, because it allows us to observe more lending relationships and the borrower's economic activity history.

We complement the administrative data with the Continuous Agricultural and Livestock Area and Production Survey (ESPAC), conducted by Ecuador's national statistics institute (INEC). ESPAC collects information on the main agricultural and livestock activities, covering all 23 continental provinces and representative at the province level. Its finance module records whether a producer applied for a loan and whether the application was approved. We use the 2020 survey to identify agricultural producers whose loan applications were rejected. We exclude producers with farms larger than 20 hectares, since smallholders are the primary borrowers in the microcredit segment.

We also use a measure of the distance between each borrower and each cooperative's closest branch, constructed in Ontaneda (2026), which enters the empirical model as determinant of demand.[4]

Table 1 presents summary statistics for agricultural and non-agricultural loans. The mean interest rate is 22.19%, above the agricultural cap of 20.97%. The average loan size is \$5,794, and average borrower income is \$1,947 per month. Of all loans, 48% are issued to borrowers with at least secondary education, and 52% to borrowers who own their home. Loans reported as financing agricultural activities account for 30% of the sample. The mean distance between a borrower's parish and the closest branch of the originating cooperative is 11.5 km. On average, borrowers obtain credit from 1.2 cooperatives and take about two loans over the sample period. Cooperatives operate an average of 1.07 branches in each province where they operate, and a canton has roughly 14 cooperatives on average.

[4]See Ontaneda (2026) for details on the construction of this variable.

Table 1: Summary statistics

	Mean	SD	Min	Max	N
Panel A: Loans					
Interest rate (p.p.)	22.19	3.57	10.00	30.50	1,000,146
Loan amount (USD)	5,794	8,426	1	660,000	1,000,146
Income (USD)	1,947	48,854	0	45,910,823	1,000,146
High education	0.48	0.50	0.00	1.00	1,000,146
Home owner	0.52	0.50	0.00	1.00	1,000,146
Agricultural activity	0.30	0.46	0.00	1.00	1,000,146
Distance to branch (km)	11.51	13.73	0.86	462.34	1,000,146
Panel B: Borrowers					
Cooperatives per borrower	1.19	0.47	1.00	8.00	620,932
Loans per borrower	1.92	1.32	1.00	10.00	620,932
Panel C: Markets					
Branches per province	1.07	2.30	0.00	20.00	797
Lenders per canton	14.08	8.69	1.00	74.00	217

Notes: Panel A reports statistics at the loan level for cooperative originations, 2017–2020. Panel B reports one observation per borrower. Panel C reports one observation per province (branches) and per canton (lenders).

3 Descriptive Evidence

Because regulatory classification relies on loan-level reporting by the originating lender without complete enforcement, the reform allowed for noncompliance. A lender could avoid the agricultural cap by classifying an agricultural loan under a non-agricultural subsegment, or by reporting the loan as financing a non-agricultural activity. In this section we first document the effect of the interest rate cap and then the lenders' response to that. These responses include strategies they use to avoid the cap, how they price non-compliant loans, which lenders choose not to comply, and how the reform affects credit access across new vs repeat borrowers.

3.1 Effects of the Interest Rate Cap

We first show the effect of the interest rate cap on interest rates and reported agricultural lending across agricultural and non-agricultural loans. These patterns are consistent with the standard trade-off of interest rate caps between interest reduction and credit rationing.

Table 2 shows that interest rates on agricultural loans fall 0.36 percentage points after the reform relative to non-agricultural loans. This decline is consistent with the intended effect of the cap. While rates on non-agricultural loans also fall in the post reform period, it reflects the downward trend in cooperative rates over the sample period (Appendix Figure A.1).

Table 2: Interest rates on agricultural loans after the reform

	(1) Interest rate
Agriculture × Post	-0.360**
	(0.142)
Post	-0.187
	(0.159)
Lender × Agriculture FE	Yes
Month-of-year FE	Yes
Pre-reform mean, agriculture	20.906
R^2	0.559
Observations	657,073

Notes: The unit of observation is a loan between March 2018–February 2020, weighted by loan amount. The dependent variable is the annual effective interest rate (p.p.). Agriculture equals one for loans to agricultural sector. Post equals one from March 2019 onward. Standard errors clustered by lender in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

The lower rates on agricultural loans come with a contraction in reported agricultural lending (Table 3; Appendix Figure A.2). Agricultural lending volume fell by approximately 18% relative to the pre-reform period, despite no change in total lending. This pattern is consistent with lenders rationing credit in the capped sector. This fall can also

potentially explain how lenders report some agricultural loans as non-agricultural, in which case the borrower is still served but the loan no longer appears in agricultural lending. Distinguishing these explanations requires measuring noncompliance and credit access at the borrower level.

Table 3: Agricultural and total lending after the reform

	Agricultural		Total	
	(1)	(2)	(3)	(4)
	Loans	Volume	Loans	Volume
Post	-0.201***	-0.201**	0.007	0.058
	(0.075)	(0.092)	(0.029)	(0.047)
Lender FE	Yes	Yes	Yes	Yes
Month-of-year FE	Yes	Yes	Yes	Yes
Pseudo R^2	0.704	0.762	0.919	0.953
Observations	1,968	1,968	1,968	1,968

Notes: Poisson pseudo-maximum likelihood estimates. The unit of observation is lender × month, March 2018–February 2020. Post equals one from March 2019 onward. Loans is the number of originations and Volume the amount originated. Total includes agricultural and non-agricultural lending. Standard errors clustered by lender in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

At the borrower level, this decline in volume as well as loans is heterogeneous across borrower types. (Table 4). While lending to new and repeat borrowers falls by similar amounts, it does differentially for those with single vs multiple lending relationships. Agricultural loans to single-lender borrowers fall by 16%, while lending to multiple-lender borrowers falls by 24%. Multiple-lender borrowers have outside options, so cooperatives have less room to retain them. Reclassification removes loans from reported agricultural lending for both groups, but more so for single-lender borrowers (Table 5). Single-lender borrowers therefore keep more of their cooperative credit, often through

noncompliant loans, while multiple-lender borrowers lose more of it.

Table 4: Agricultural lending after the reform, by borrower type

	New vs. repeat		Single vs. multiple	
	(1)	(2)	(3)	(4)
	Loans	Volume	Loans	Volume
Post	-0.223***	-0.239**	-0.178**	-0.179*
	(0.083)	(0.103)	(0.074)	(0.092)
Post × New	0.047	0.074		
	(0.051)	(0.061)		
Post × Multiple			-0.097***	-0.080**
			(0.029)	(0.031)
Lender × group FE	Yes	Yes	Yes	Yes
Month-of-year FE	Yes	Yes	Yes	Yes
Pseudo R^2	0.703	0.771	0.749	0.795
Observations	3,936	3,936	3,936	3,936

Notes: Poisson pseudo-maximum likelihood estimates on agricultural loans. The unit of observation is lender × month × borrower group, March 2018–February 2020. Post equals one from March 2019 onward. New borrowers have no cooperative loan in the previous 24 months. Multiple-lender borrowers are observed borrowing from more than one cooperative. Standard errors clustered by lender in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

3.2 Two Noncompliance Strategies

We define noncompliance as a loan extended to an agricultural borrower at an interest rate above the regulatory cap. In our current context, noncompliance can be operationalized in two ways: (a) *Subsegment Misclassification:* By classifying an agricultural activity under a non-agricultural regulatory subsegment; (b) *Activity Reclassification:* By reporting

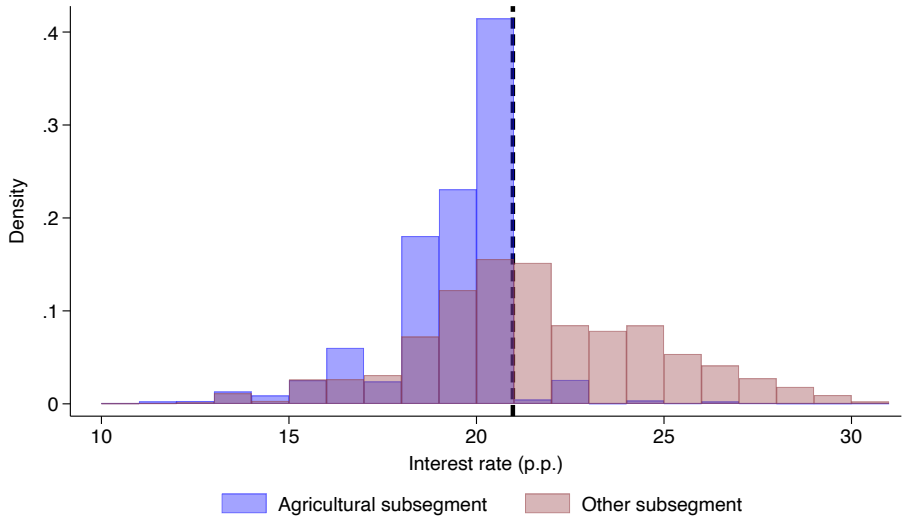
an agricultural borrower's loan under a non-agricultural economic activity.

3.2.1 Subsegment Misclassification

Since both the activity code and the subsegment classification are reported by the originating lender, this form of noncompliance is directly observable in the data.

Figure 2 shows the distribution of interest rates on loans that finance agricultural activities after the reform. Loans that lenders classify in the agricultural subsegment cluster at or just below the cap. Loans that finance the same activities but are classified in other subsegments spread well above it. This gap reveals the existence of subsegment misclassification.

Figure 2: Post-Reform Distribution of Interest Rates by Regulatory Subsegment



Note: This figure shows the distribution of interest rates for loans financing agricultural activities, separated by regulatory subsegment. The dashed vertical line marks the agricultural interest rate cap.

3.2.2 Activity Reclassification

The second strategy is harder to observe directly. Because the true economic activity of a borrower is latent, we cannot identify this form of noncompliance on a loan-by-loan basis. Instead, we exploit the panel structure of the data and focus on borrowers with repeat loans over the sample period.

We define an *activity switch* as an instance in which a borrower's reported economic activity differs across two consecutive loans. Under the null hypothesis of truthful reporting, the rate of switching and its direction (into or out of agriculture) should be stable over time. Thus, *activity reclassification* is the excess post-reform rate of switching out of agriculture relative to the pre-reform baseline. The identifying assumption is that, in the absence of the regulation, the direction of reported activity changes would be stable over time.

Table 5: Sector switching after the reform

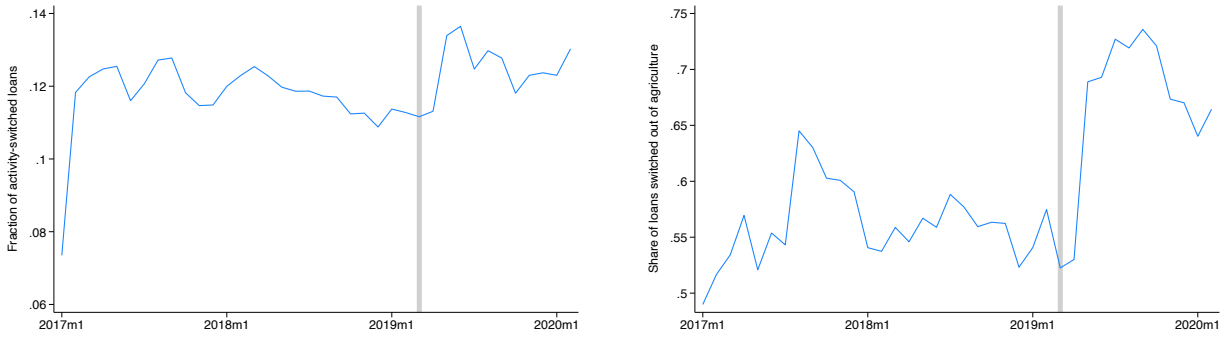
	Any Switch	Switch out of Agriculture		
	(1) Repeat loans	(2) Switchers	(3) Single rel.	(4) Multiple rel.
Post	-0.018***	0.119***	0.132***	0.106***
	(0.001)	(0.003)	(0.004)	(0.005)
Pre-reform mean	0.177	0.570	0.594	0.541
R^2	0.001	0.013	0.017	0.011
Observations	534,717	90,758	48,955	41,803

Notes: The unit of observation is a repeat loan, i.e. a borrower's second or later loan, between March 2017-February 2020. In column 1 the dependent variable equals one if the loan's sector (agricultural or non-agricultural) differs from that of the borrower's previous loan. In columns 2-4 the sample is repeat loans that switch sector, and the dependent variable equals one if the switch is out of agriculture. Single (multiple) relationship borrowers borrow from one (more than one) cooperative. Post equals one from March 2019 onward. Robust standard errors in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

Table 5 shows evidence of activity switching after the reform. For repeat borrowers, switching does not happen more often after the reform. While the overall switching rate falls slightly, what changes is the direction of switching. After the reform, a larger share of switches move out of agriculture, an increase of about 21% relative to the pre-reform

share. Panel (a) in figure 3 shows that the direction of switching is stable before the reform. The shift toward switching out of agriculture therefore reflects activity reclassification rather than genuine changes in borrowers' activities (Panel (b) Figure 3).

Figure 3: Activity-Switched Loans, 2017–2020



(a) Rate of activity switching

(b) Share of loans switched out of agriculture

Note: The vertical line marks the reform date. Left panel shows the fraction of loans with a different reported activity than the borrower's previous loan. Right panel shows the share of switches directed out of agriculture.

3.3 Extent of Noncompliance

Combining both strategies, we estimate that about 37 percent of agricultural loans issued after the reform did not comply with the interest rate cap (Table 6). Because reclassified loans are reported as non-agricultural, they are missing from reported agricultural lending, so we add them back to obtain the total number of agricultural loans. Of the 85,350 agricultural loans after this adjustment, 28,261 (33 percent) are subsegment misclassified and an estimated 3,071 (4 percent) are reclassified. We measure subsegment misclassification directly. For activity reclassification, we count only the excess of post-reform switching out of agriculture over its pre-reform average, and we assume these loans carry rates above the cap as often as other loans leaving agriculture. Because reclassification is observed only for repeat borrowers and the pre-reform average exceeds its most recent level, this estimate is likely conservative.

Table 6: Extent of noncompliance after the reform

Panel A: Subsegment misclassification	
Loans with agricultural activity	80,896
Other subsegment, above cap	28,261
Share of loans with agricultural activity (%)	34.9
Panel B: Activity reclassification	
Repeat loans, previous loan agricultural	63,195
Observed share leaving agriculture (%)	34.1
Pre-reform average share leaving agriculture (%)	27.1
Excess loans leaving agriculture	4,454
Share of leavers above cap (%)	68.9
Reclassified, above cap	3,071
Share of repeat loans (%)	4.9
Panel C: Share of agricultural loans, including reclassified	
Agricultural loans including excess leavers	85,350
Subsegment misclassification (%)	33.1
Activity reclassification (%)	3.6
Total noncompliance (%)	36.7

Notes: The post-reform period is March 2019-February 2020. Above cap means an interest rate above 20.97%. Panel B compares the share of repeat loans leaving agriculture with its average over the pre-reform period, March 2017-February 2019. Excess loans are assumed to be above the cap at the same rate as all post-reform leavers. Activity reclassification is observed only for repeat loans.

3.4 Pricing Noncompliant Loans

Lenders price compliant and noncompliant loans differently. Under subsegment misclassification, loans in the agricultural subsegment cluster at or just below the cap, while those in other subsegments spread well above it (Figure 2). Activity reclassification shows the same pattern (Table 7. Figure 4). Relative to loans that remain non-agricultural, rates on loans that stay in agriculture fall after the reform. However, rates on loans reclassified out of agriculture do not fall, and are about 0.5 percentage points higher than rates on loans that stay in agriculture. Thus, interest rates on loans that remain in agriculture falls, while for those reclassified out of agriculture it does not.

Table 7: Interest rates on switched loans after the reform

	(1) Interest rate
Stayed in Agriculture × Post	-0.334*
	(0.178)
Left Agriculture × Post	0.185
	(0.142)
Entered Agriculture × Post	-0.183
	(0.146)
Left minus Stayed in Agriculture	0.519**
	(0.213)
Group FE	Yes
Lender FE	Yes
Month FE	Yes
R^2	0.526
Observations	534,717

Notes: The unit of observation is a repeat loan, i.e. a borrower's second or later loan, between March 2017-February 2020, weighted by loan amount. The dependent variable is the annual effective interest rate (p.p.). Groups are defined by the sector of the loan and of the borrower's previous loan. The omitted group is loans that stayed in the non-agricultural sector. Post equals one from March 2019 onward. Standard errors clustered by lender in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

3.5 Which Lenders Do Not Comply

Noncompliance is primarily a lender-level choice. Lender fixed effects explain far more of the variation in both strategies than borrower or location characteristics (Appendix Figure B.1). Table B.1 relates noncompliance to three lender and market characteristics that enter the model. The unit of observation is a lender-market pair. Appendix B reports the underlying figures.

Pre-reform interest rates: A one-standard-deviation increase in the pre-reform interest rate is associated with a 24.6 percentage-point increase in the subsegment misclassifica-

Figure 4: Interest Rates on Activity-Switched Loans, 2017–2020



Note: Loans switched out of agriculture maintain higher interest rates after the reform relative to loans switched into agriculture or unswitched loans. The vertical line marks the reform date.

tion rate (column 1) and a 3.4 percentage-point increase in the activity reclassification rate (column 2), both statistically significant (Appendix Figure B.2). Lenders whose optimal rates sit farther above the cap lose more from complying.

Number of lenders: Subsegment misclassification is more common in markets with more lenders (Appendix Figure B.4), where competition makes it harder to keep borrowers at rates below the cap.

Market share: Lenders with larger pre-reform market shares are less likely to evade (Appendix Figure B.3), consistent with greater regulatory scrutiny or with enough market power to retain borrowers under compliance.

The two strategies are positively correlated across lenders (Appendix Figure B.5), so lenders that evade through one channel also evade through the other.

Together, these patterns show that lenders respond to the cap along three margins: whether

to serve a borrower, whether to comply, and how to evade. These choices depend on lender pricing, market structure, and the borrower's lending relationships. Reported lending cannot separate lost access from reclassification. The next section develops a model of these choices to quantify how noncompliance shapes the effects of the cap.

4 Empirical Model

We model the market for agricultural microcredit served by cooperatives. Lenders decide whether to make an offer and which contract to offer, and borrowers decide whether to accept or decline it. We distinguish between repeat borrowers (i.e., who have borrowed before) and new borrowers.

4.1 Regulation

The interest rate regulation defines regulatory subsegments, denoted by s. Before the reform, agricultural and non-agricultural loans are classified in the same subsegments. The reform creates a new agricultural subsegment, whose rate is subject to the agricultural cap $\bar{r}$, while the pre-existing subsegments remain available for non-agricultural loans.

A lender complies by classifying an agricultural loan in the agricultural subsegment. It does not comply by classifying it in a pre-existing subsegment, either recorded as agricultural (subsegment misclassification) or as a non-agricultural activity (activity reclassification).

4.2 Demand

The indirect utility that agricultural borrower i derives from accepting a loan offer extended by lender j is given by:

$$u_{ij} = \alpha r_{ij} + \beta_0 Z_i + \beta_1 X_{ij} + \xi_j + \zeta_{ij} + \epsilon_{ij}, \tag{1}$$

where r_{ij} is the interest rate, Z_i are borrower-specific characteristics that do not vary across lenders, X_{ij} is a vector of characteristics that vary at the borrower-lender level (distance to the closest branch, branch network, prior relationships), and ξ_j captures the lender's unobserved characteristics. The term ζ_{ij} is an unobserved borrower-lender demand shock, while the idiosyncratic taste shock ϵ_{ij} is assumed to be distributed EVT1. The utility of the outside option, not borrowing from a cooperative, is given by $u_{i0} = \epsilon_{i0}$, with the deterministic component normalized to zero.

The probability that repeat borrower i accepts an offer made by lender j at rate r is:

$$q_{ij}(r) = \frac{\exp\left(\tilde{u}_{ij}(r)\right)}{1 + \sum_{j' \in \mathcal{J}_i} \exp\left(\tilde{u}_{ij'}\right)}, \tag{2}$$

where $\mathcal{J}_i$ is the set of lenders available to borrower i, and $\tilde{u}_{ij}$ denotes the deterministic component of the indirect utility.

We assume that new borrowers are credit constrained and accept any offer they receive ($q_{ij}(r) = 1$). Their access depends only on the lender's offer decision.

4.3 Supply

On the supply side, lender j chooses interest rates and decides for each borrower it is matched with, whether to make an offer and which contract to offer.

Interest rates. The marginal cost of lending to borrower i is $mc_{ij} = \kappa_{js} + \delta_i$, where κ_{js} is lender j's cost of lending in subsegment s and δ_i is a borrower-specific deviation with mean zero within the subsegment, reflecting, for instance, risk. Lender j sets a base rate r_{js} for each subsegment and offers borrower i the rate $r_{ij} = r_{js} + \delta_i$, where the cost deviation

passes through one-for-one to the rate. Therefore, the margin $r_{ij} - mc_{ij} = r_{js} - \kappa_{js}$ is the same for all borrowers in the subsegment.

Before the reform, the lender chooses the base rate to maximize expected variable profits over all potential borrowers in the subsegment:

$$\max_{r_{js}} \sum_{i \in s} q_{ij}(r_{js} + \delta_i) \left(r_{js} - \kappa_{js}\right). \tag{3}$$

The first-order condition gives:

$$r_{js} = \kappa_{js} - \frac{\sum_{i \in s} q_{ij}}{\alpha \sum_{i \in s} q_{ij} \left(1 - q_{ij}\right)}. \tag{4}$$

After the reform, a lender can classify an agricultural loan in the agricultural subsegment at the compliant rate $r_{ij}^C = \min\{r_{js} + \delta_i, \bar{r}\}$, or in an pre-existing subsegment at the non-compliant rate $r_{ij}^N = r_{js} + \delta_i$. We assume that the compliant rate is the pre-reform rate truncated at the cap, and noncompliant agricultural loans are priced like any loan in the same subsegment.

Lender Contract Choice. After the reform, the possible contracts are $\{C, N1, N2, 0\}$: a compliant agricultural loan (C), a noncompliant loan classified under a non-agricultural subsegment ($N1$, subsegment misclassification), a noncompliant loan reported as a non-agricultural activity ($N2$, activity reclassification), or no offer (0). Noncompliance is only available when $r_{js} + \delta_i > \bar{r}$. Before the reform, there is no cap and no noncompliance, so lenders choose only between offering a loan at the base rate and making no offer. Lender j's expected payoff from offering contract k to borrower i in period t is:

$$V_{ijt}^k = \lambda \pi_{ijt}^k - F_{it} - \eta_{ij}^k + \varepsilon_{ijt}^k, \qquad k \in \{C, N1, N2\} \tag{5}$$

where λ measures the sensitivity of lenders' contract choices to expected variable profits, $\pi_{ij}^k = q_{ij}\left(r_{ij}^k\right)\left(r_{ij}^k - mc_{ij}\right)$. F_{it} is the cost of making an offer, η_{ij}^k represents the cost of non-

compliance ($\eta_{ij}^C = 0$), and ε_{imjt}^k are idiosyncratic shocks distributed EVT1. We normalize the deterministic component of not offering to zero, $V_{ijt}^0 = \varepsilon_{ijt}^0$. The lender chooses the contract that delivers the highest payoff.

We specify the offer cost as $F_{it} = F + F_t + F_i$, where F is a baseline cost. F_t captures the change in the opportunity cost of agricultural lending after the reform, reflecting that extending offers in the non-agricultural sector becomes relatively more profitable. $F_t = 0$ before the reform and $F_t = F_{post}$ after. F_i captures the additional cost of serving a new borrower, where $F_i = F_{new}$ for new borrowers and zero for repeat borrowers.

The noncompliance cost is:

$$\eta_{ij}^k = \rho_1 \frac{1}{N_m} + \rho_2 MS_{mj} + \tau^k + \tau_i, \qquad k \in \{N1, N2\}, \tag{6}$$

where m is the market of borrower i, N_m is the number of cooperatives lending in market m, and MS_{mj} is lender j's market share in market m. τ^k is a cost specific to the type of noncompliance, common to all lenders, and τ_i shifts the cost for new borrowers; that is, $\tau_i = \tau_{new}$ for new borrowers and zero for repeat borrowers. We assume that the noncompliance cost does not affect pricing.

Given the distribution of the shocks, the probability that lender j offers contract k to borrower i is:

$$P_{ij}^k = \frac{\exp\left(\bar{V}_{ij}^k\right)}{1 + \sum_{k' \in \mathcal{K}_{ij}} \exp\left(\bar{V}_{ij}^{k'}\right)}, \qquad k \in \mathcal{K}_{ij}, \tag{7}$$

where $\bar{V}_{ij}^k$ is the deterministic part of the payoff and $\mathcal{K}_{ij}$ is the set of contracts available.

Combining offers and acceptance, the probability of observing a contract k is $P_{ij}^k q_{ij}(r_{ij}^k)$, and the probability of observing no loan is $P_{ij}^0 + \sum_{k \in \mathcal{K}_{ij}} P_{ij}^k \left(1 - q_{ij}(r_{ij}^k)\right)$.

5 Estimation and results

In this section we detail the estimation strategy and present the estimation results.

5.1 Estimation and identification

We estimate the model in two steps. First, we estimate demand using repeat borrowers in the pre-reform period. Second, conditional on demand, we recover the cost of lending in each subsegment and estimate the parameters of the lenders' payoff using the pre- and post-reform periods jointly.

Preliminaries. We define a pre-reform period (from March 2018 to February 2019), and a post-reform period (from March 2019 to February 2020). For each period, we observe the borrowers' history with cooperatives over the preceding fourteen months, which we treat as a history window.

Estimation requires a definition of choice sets along with the interest rates for each lender within the choice set. We distinguish between repeat and new borrowers. Repeat borrowers are borrowers with at least one agricultural loan from a cooperative in the history window. For each repeat borrower, the set of available lenders $\mathcal{J}_i$ is the set of cooperatives the borrower has borrowed from in the history window. We classify repeat borrowers as agricultural based on their loans in the history window, which are not affected by the reform.

New borrowers are borrowers with no loan from a cooperative in the history window. Because we cannot use past loans to classify new borrowers as agricultural, we classify them using the loan they take in the period. This implies that we cannot observe new borrowers whose activity was reclassified as non-agricultural, and they are not part of the sample. In the administrative data we only observe new borrowers who received a loan. We complement them with farmers in the agricultural survey who applied for credit

and were rejected, treating them as new borrowers who did not receive an offer. Since the survey does not identify the lender, we assume that the rejected applicant's choice set only includes the closest cooperative that lends to new borrowers in the province.

We include all repeat borrowers in both periods. The outcome is the lender of the borrower's first loan from a cooperative in the period, or the outside option if she does not borrow during the period. We control for the number of loans still outstanding at the start of the period and the months since the last loan. We use new borrowers and rejected applicants only in the post-reform period in the supply estimation.

We observe the interest rate only for the loan the borrower takes. Following Crawford et al. (2018), we predict the interest rate the borrower would have paid at each lender in $\mathcal{J}_i$ with a regression of observed rates on the borrower's last rate with the lender, the loan amount and maturity, lender-by-subsegment and month fixed effects, and borrower characteristics. These predicted rates are used to estimate demand. We explain the rates that enter the supply estimation below.

Demand. We estimate demand by maximum likelihood on the choice occasions of repeat borrowers in the pre-reform period following the specification in equation (1). The borrower-lender characteristics X_{ij} are the number of previous loans with the lender, the log distance to the closest branch, and the log number of branches in the province. The borrower characteristics Z_i are log income, education, home ownership, months since the last loan, the number of outstanding loans, the log amount of the last loan, crop type fixed effects, and fixed effects for the total number of loans in the history window. The specification includes lender-fixed effects, ξ_j.

We address endogeneity arising from the potential correlation between interest rates and the borrower-lender demand shock ζ_{ij} with a control function approach (Petrin and Train, 2010). In a first stage, we regress the interest rate on an instrument and on all the covariates in borrower utility. The instrument is the lender's average rate in other provinces in the same month and subsegment, reflecting changes in the lender's rates that are com-

mon across its markets, such as funding costs. It is excluded from borrower utility under the assumption that demand shocks are local. We then assume that $E\left[\zeta_{ij} \mid r_{ij}, v_{ij}\right] = \varphi\, v_{ij}$, where v_{ij} is the first-stage residual and φ is a parameter to be estimated. We substitute ζ_{ij} with $\varphi\, v_{ij}$ in equation (1).

Supply. Estimating supply-side parameters requires two interest rates for each potential borrower after the reform: a compliant and a noncompliant rate. While we only observe the interest rate of the chosen contract, both rates depend on the base rate r_{js} and the borrower deviation δ_i. We measure the post-reform base rate as the lender's average rate on non-agricultural loans in the same subsegment and month.

For borrowers with a noncompliant loan, the noncompliant rate is the observed rate and the corresponding compliant rate equals the cap. For borrowers with a compliant loan below the cap, we interpret the observed rate as satisfying equation (4). Since it is below the cap, the cap does not bind and noncompliance yields no gain, so we exclude it from the lender's choice set. For borrowers we observe with compliant loans at the cap and borrowers without a loan, the deviation is not observed. We approximate it by recovering δ_i from pre-reform loans as the difference between the observed rate and the base rate and integrating over the empirical distribution of δ.[5] For compliant loans at the cap, we keep only the draws for which the base rate plus the deviation is at or above the cap. We integrate over the entire distribution for borrowers without a loan.

Let $y_{it} \in \{0, C, N1, N2\}$ denote the observed outcome let θ denote the supply parameters. The likelihood of the observed outcome is

$$L_{it}(\theta) = \sum_{k \in \mathcal{K}_{ijt}} \mathbb{1}\{y_{it} = k\}\, P^k_{ijt}(\theta)\, q_{ij}\left(r^k_{ijt}\right) + \mathbb{1}\{y_{it} = 0\} \left[P^0_{ijt}(\theta) + \sum_{k \in \mathcal{K}_{ijt}} P^k_{ijt}(\theta) \left(1 - q_{ij}\left(r^k_{ijt}\right)\right) \right]. \tag{8}$$

We estimate θ by maximizing the weighted log-likelihood over the pre- and post-reform

[5] We use pre-reform loans because observed rates are not truncated by the cap.

periods,

$$\max_{\theta} \sum_{t} \sum_{i} w_i \log L_{it}(\theta), \tag{9}$$

with weight w_i equal to one for borrowers in the administrative data and to the survey weight for rejected applicants.

The sample is composed of repeat borrowers who borrowed from a single cooperative in the history window, with at least two loans, and new borrowers after the reform. For repeat borrowers with a single lender, $\mathcal{J}_i$ contains only the lender they are matched with. We normalize the baseline offer cost F so that, before the reform, every repeat borrower receives an offer. This is consistent with the demand step, which treats every pre-reform borrower without a loan as having declined an offer. We evaluate the acceptance probability at the estimated demand parameters. Because the samples used to estimate supply and demand are not the same, we shift the deterministic utility of repeat borrowers by a constant chosen so that, before the reform and with every borrower receiving an offer, the predicted share of borrowers with a loan equals the observed share. The same shift applies after the reform.

Identification. The sensitivity to variable profits λ is identified from how contract choices vary with variable profits across borrowers. The noncompliance costs τ^{N1} and τ^{N2} are identified from the shares of each type of noncompliance among borrowers for whom noncompliance yields a gain, and ρ_1 and ρ_2 from how noncompliance varies with market structure. Market structure is measured before the reform. τ_{new} is identified from the share of noncompliant loans among new borrowers. The post-reform offer cost F_{post} is identified from the change in repeat borrowers without a loan after the reform, beyond what demand predicts at post-reform rates. The offer cost of new borrowers F_{new} is identified from the share of new borrowers who receive an offer, that is, new borrowers in the administrative data relative to rejected applicants in the survey.

5.2 Results

Table 8 presents the results of the demand estimation. As expected, borrowers dislike higher interest rates. The implied median elasticity of demand with respect to the interest rate is -2.36. The control function residual is positively correlated with demand, suggesting that interest rates are positively correlated with unobserved demand shocks. The relationship with the lender is an important determinant of demand. Each previous loan with the lender increases the utility of borrowing from it. Borrowers are more likely to accept offers from lenders geographically closer. The number of branches in the province has a small, imprecisely estimated effect. Among borrower characteristics, borrowers with higher income and home owners are more likely to borrow. Borrowers with more outstanding loans and more months since their last loan are also more likely to borrow again, while borrowers whose last loan was larger are less likely to accept a loan offer.

Table 8: Demand estimates

	Estimate	SE
Interest rate	-0.199	0.012
Control function residual	0.149	0.013
Previous loans with the lender	0.442	0.026
Log distance to branch	-0.059	0.013
Log branches in province	-0.033	0.019
Log income	0.140	0.011
Higher education	-0.068	0.017
Home owner	0.105	0.017
Months since last loan	0.025	0.002
Outstanding loans	0.335	0.021
Log last loan amount	-0.543	0.012
Lender fixed effects	Yes	
Crop type fixed effects	Yes	
Number of previous loans fixed effects	Yes	

Notes: This table reports the estimated demand parameters along with standard errors. The control function residual comes from a first stage on the leave-out rate instrument.

Table 9 reports the supply estimates. The profit weight implies $\lambda = \exp(-2.139) \approx 0.12$. Noncompliance costs depend on market structure. The cost is higher in markets with

fewer lenders and lower for lenders with a larger market share. The constants of both types of noncompliance are negative, which implies that noncompliance is preferred to compliance for reasons beyond the difference in variable profits. Noncompliance is more costly for new borrowers. The offer cost after the reform almost offsets the normalized baseline, which implies that the lender no longer extends an offer to every repeat borrower. Serving a new borrower is more costly than serving a repeat borrower.

Table 9: Supply estimates

	Estimate	SE
Profit weight (log)	-2.139	0.038
Inverse number of cooperatives in the canton	5.838	0.270
Lender market share	-1.081	0.067
Noncompliance cost, subsegment misclassification	-2.163	0.077
Noncompliance cost, activity reclassification	-1.981	0.077
Noncompliance cost shift, new borrowers	1.565	0.072
Offer cost after the reform	29.391	0.071
Offer cost, new borrowers	1.167	0.068

Notes: Pre- and post-reform supply estimated jointly, in utility units. Before the reform, all clients with a history receive an offer (offer cost normalized); the offer cost after the reform is common to all clients and captures the higher opportunity cost of agricultural lending. New clients are new borrowers and rejected applicants from the agricultural survey, weighted by survey weights. Standard errors are preliminary: they ignore first-stage estimation and the survey weights.

Table 10 presents statistics on the model fit for repeat and new borrowers in the post-reform period. Generally, the model fits well the share of borrowers with compliant and noncompliant loans as well as the share of farmers not borrowing. The model predicts that the share of repeat borrowers receiving an offer is substantially larger than the corresponding share among new borrowers.

6 The distributional effects of noncompliance

Using the estimated model, we quantify the distributional effects of noncompliance by simulating a counterfactual scenario under full compliance. In this counterfactual, lenders

Table 10: Model fit

	Repeat borrowers		New borrowers	
	Observed	Predicted	Observed	Predicted
No loan	0.480	0.498	0.293	0.285
Compliant loan	0.280	0.276	0.362	0.350
Noncompliant: subsegment misclassification	0.131	0.123		
Noncompliant: activity reclassification	0.109	0.103		
Noncompliant: any	0.240	0.226	0.344	0.364
Noncompliant share of loans	0.462	0.451	0.487	0.511
Offer rate (predicted)		0.898		0.715

Notes: Post-reform period. New clients are new borrowers and rejected applicants from the agricultural survey, weighted by survey weights. The two types of noncompliance are not distinguished for new clients.

choose whether or not to offer a compliant loan. We hold fixed the base rates, the borrower deviations, and the post-reform offer cost, so the compliant rate of each borrower is the same as in the baseline with noncompliance. We also impose that lenders do not offer contracts with negative expected profits. We report the share of potential borrowers with a loan, the average interest rate of the loans, the offer rate, consumer surplus, and lender expected profits per potential borrower. Consumer surplus is measured in interest-rate points.

Table 11 reports the results. Full compliance reduces access to credit, and the effect is much larger for new borrowers. When noncompliance is not available, lenders offer the compliant contract to some of the borrowers who would have received a noncompliant contract and do not make an offer to the rest. The offer rate falls by 15 percentage points (17 percent) for repeat borrowers and by 18 percentage points (25 percent) for new borrowers.

For repeat borrowers, the reduction in offers translates into a small reduction in borrowing. Since borrowers who keep an offer pay at most the cap, the average interest rate falls by 1.2 percentage points, which increases the probability of accepting an offer, offsetting most of the reduction in offers. As a result, consumer surplus of repeat borrowers increases by 1% because the gains of those who borrow at a lower rate exceed the losses of those who no longer receive an offer.

For new borrowers, the reduction in offers fully translates into a reduction in borrowing due to the assumption of inelastic demand. The share of new borrowers with a loan decreases 18 percentage points. Although the average interest rate falls by 1.7 percentage points among new borrowers, their consumer surplus decreases by 9%.

Table 11: Market outcomes and welfare under full compliance

	Repeat borrowers			New borrowers		
	Baseline	Full compliance	Change	Baseline	Full compliance	Change
Share with a loan	0.502	0.479	-0.022	0.714	0.533	-0.182
Noncompliant share of loans	0.451	0.000	-0.451	0.511	0.000	-0.511
Average rate (pp)	21.075	19.907	-1.169	21.490	19.764	-1.726
Consumer surplus (pp)	4.647	4.692	0.045	2.128	1.940	-0.188
Lender profit (pp)	5.144	4.421	-0.723	5.355	3.518	-1.838
Offer rate	0.898	0.744	-0.154	0.714	0.533	-0.182

Notes: Outcomes per potential borrower at the post-reform cap (20.97 percent). Full compliance removes the noncompliant contracts; posted rates and the post-reform offer cost are held fixed. New clients are weighted by survey weights. Consumer surplus is the logit inclusive value and lender profit is expected profit per potential borrower, both in percentage points of the rate. Options with negative expected profit are not offered.

Lender profits fall for both types of borrowers, but more so among new borrowers because the reduction in the prices they pay and the extent of borrowing is larger. Overall, noncompliance allows lenders to serve borrowers for whom lending at the cap is not profitable enough. New borrowers are the group most affected by enforcing the cap.

7 Conclusion

We study how noncompliance affects targeted regulation, using a reform that introduced an interest rate cap on agricultural loans in Ecuador. We find that, after the reform, interest rates on agricultural loans fell and that agricultural lending contracted. These patterns are consistent with the trade-off between borrower protection and credit access associated with interest rate caps. However, reported outcomes understate how lenders responded. We document that lenders avoided the cap in two ways: by classifying agricultural loans under non-agricultural subsegments, and by reporting agricultural loans as financing non-agricultural activities. Noncompliance appears to be primarily a lender-level decision. It is more common among lenders with higher pre-reform interest rates and less

common among lenders with larger market shares.

We estimate an empirical model and use it to understand the effects of noncompliance. We find that noncompliance mainly helped new borrowers maintain credit access, while the effects of lower interest rates and credit rationing offset each other for repeat borrowers. Our results suggest that when regulatory implementation depends on how regulated firms classify their own transactions, classification itself becomes a margin of response. While noncompliance reduces the price benefits of the regulation, it can also mitigate its adverse consequences.

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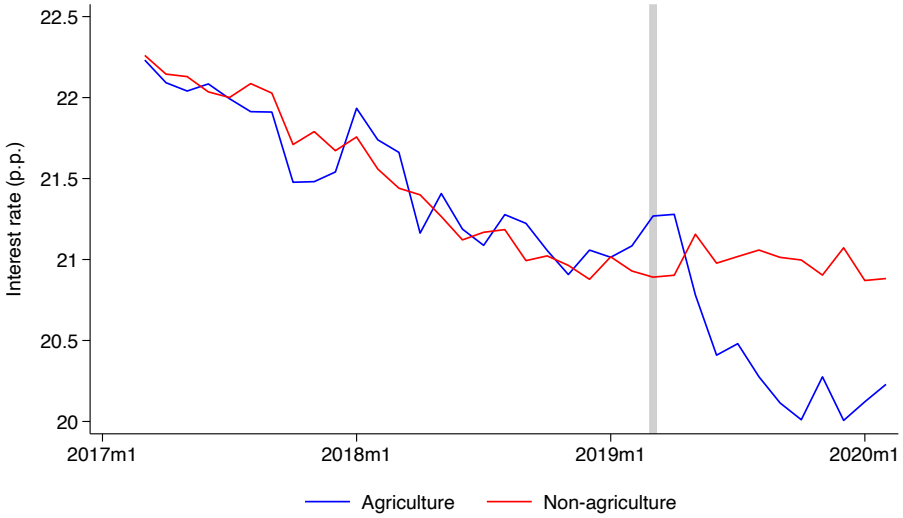
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Appendix

A Additional tables and figures

Figure A.1: Mean Effective Interest Rates by Sector



Note: This figure shows the mean effective interest rates for agricultural and non-agricultural loans from 2017-2020. The vertical line marks the date of the reform. Interest rates are measured in percentage points.

Table A.1: Subsegment misclassification, new and repeat borrowers

	Repeat	New	Difference	Within lender
Other subsegment, above cap	0.364	0.334	-0.030	-0.018
			(0.044)	(0.015)
Other subsegment, any rate	0.628	0.672	0.044	0.005
			(0.055)	(0.004)
Loans	42,305	38,591	80,896	80,896

Notes: The unit of observation is a loan with an agricultural activity code between March 2019-February 2020. Other subsegment means a non-agricultural regulatory subsegment, and above cap means an interest rate above 20.97%. New borrowers have no cooperative loan in the previous 24 months. Within lender includes lender and month-of-year fixed effects. Standard errors clustered by lender in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

Figure A.2: Credit Volumes for Agricultural and Total Lending



Note: Credit volumes for agricultural and total lending across pre-reform and post-reform periods. Credit volume is measured in USD millions.

Table A.2: Interest rates on agricultural loans after the reform, by borrower type

	(1) New vs. repeat	(2) Single vs. multiple
Post	-0.747***	-0.567***
	(0.196)	(0.191)
Post × New	0.331**	
	(0.148)	
Post × Multiple		-0.013
		(0.117)
Lender × group FE	Yes	Yes
Month-of-year FE	Yes	Yes
R^2	0.588	0.584
Observations	179,349	179,349

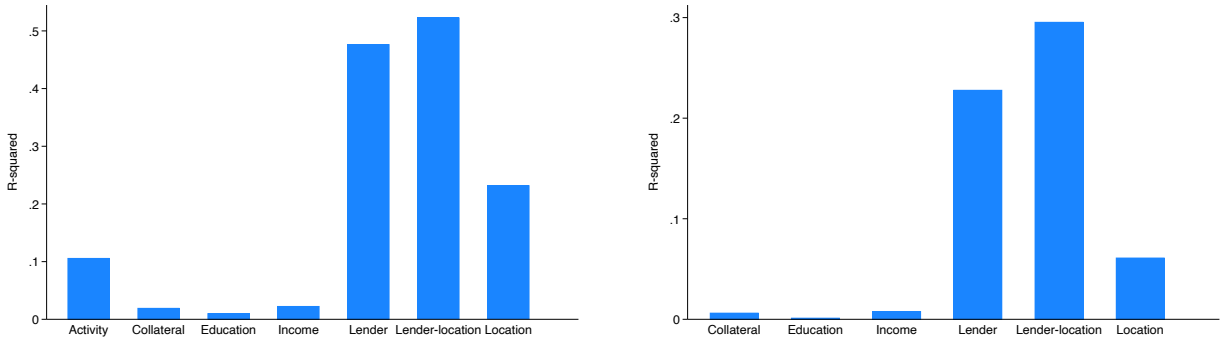
Notes: The unit of observation is an agricultural loan, March 2018–February 2020, weighted by loan amount. The dependent variable is the annual effective interest rate (p.p.). Post equals one from March 2019 onward. New borrowers have no cooperative loan in the previous 24 months. Multiple-lender borrowers are observed borrowing from more than one cooperative. Standard errors clustered by lender in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

B Stylized Facts about Noncompliance

To characterize the determinants of lender noncompliance, we examine cross-lender correlates of both strategies. Several patterns emerge consistently across subsegment misclassification and activity reclassification.

Lender variation. Variation at the lender level accounts for the preponderance of variation in both noncompliance measures. As shown in Figure B.1, R-squared values from regressions that include only lender fixed effects substantially exceed those from regressions with only borrower or location characteristics, indicating that the decision to evade is primarily a lender-level choice.

Figure B.1: Predictors of Regulatory Noncompliance



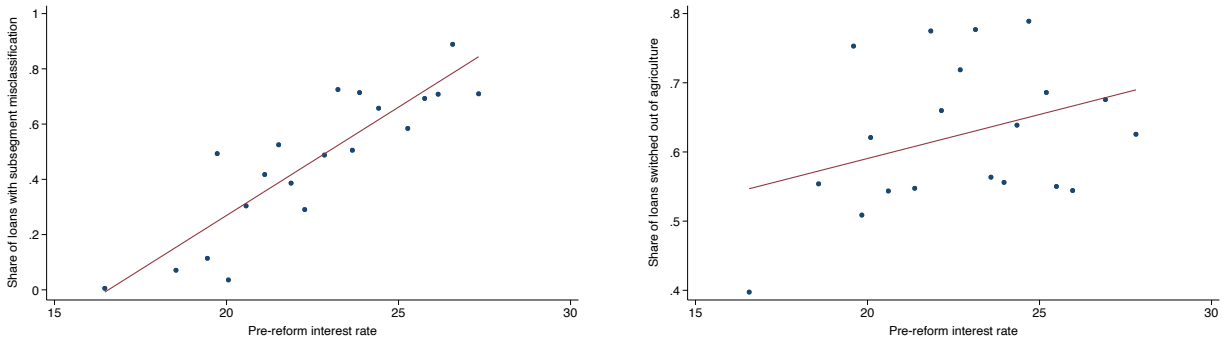
(a) Subsegment misclassification

(b) Activity reclassification

Note:This figure has predictive R-squared for each group of observables for explaining noncompliance. Lender-level variation dominates borrower and location characteristics for both strategies.

Pre-reform interest rates. Lenders with higher pre-reform interest rates are significantly more likely to engage in noncompliance after the reform, as shown in Figure B.2. This is consistent with a mechanism in which lenders whose optimal rates were farther above the cap faced greater profit pressure to evade.

Figure B.2: Noncompliance Rates and Pre-Reform Interest Rates



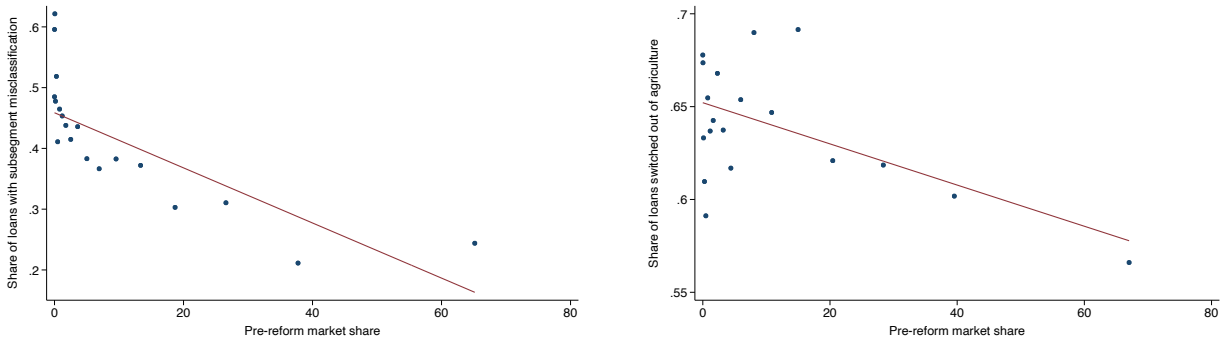
(a) Subsegment misclassification

(b) Activity reclassification

Note: Noncompliance rates against pre-reform interest rates, by lender. Both strategies are increasing in the lender's pre-reform rate.

Pre-reform market share. Lenders with higher pre-reform market shares are less likely to engage in noncompliance, as shown in Figure B.3. This pattern is consistent with dominant lenders facing greater regulatory scrutiny or reputational costs, or alternatively with lenders having sufficient market power to maintain borrower relationships under compliance.

Figure B.3: Noncompliance Rates and Pre-Reform Market Share



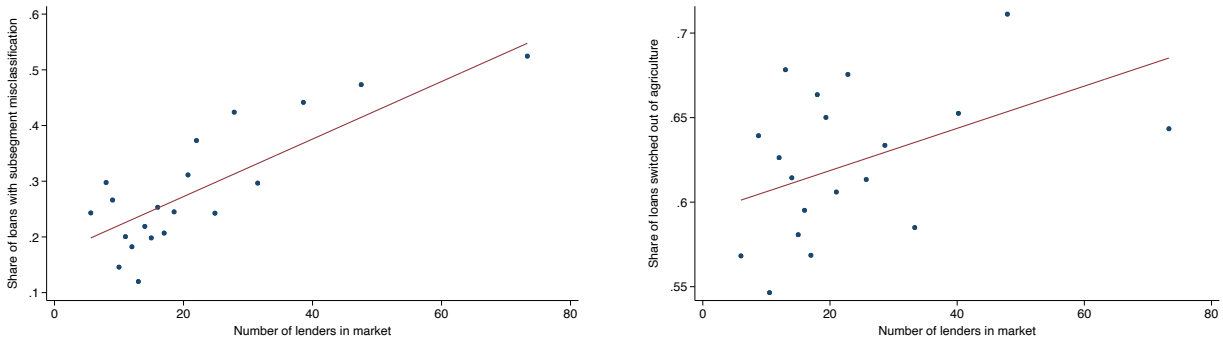
(a) Subsegment misclassification

(b) Activity reclassification

Note: Noncompliance rates against pre-reform market share, by lender. Both strategies are decreasing in market share.

Market competition. Markets with more lenders exhibit higher rates of noncompliance, as shown in Figure B.4. Competitive pressure may induce evasion as lenders seek to retain customers at non-compliant rates.

Figure B.4: Noncompliance Rates and Number of Lenders in the Market



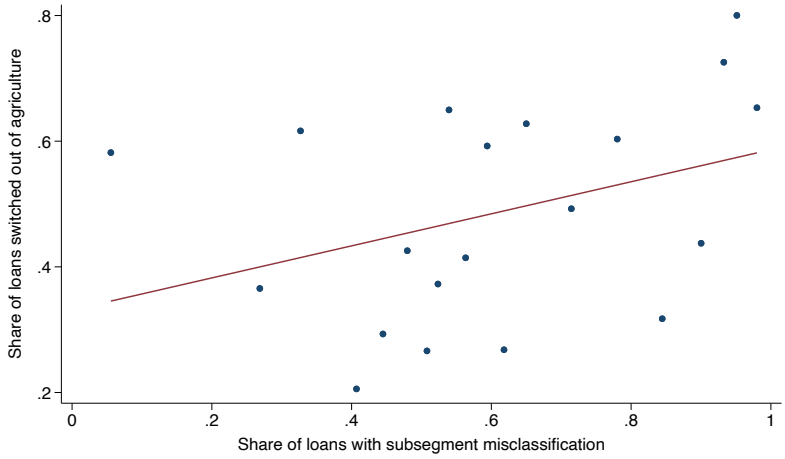
(a) Subsegment misclassification

(b) Activity reclassification

Note: Noncompliance rates against the number of lenders in the market. Subsegment misclassification is increasing in the number of lenders.

Correlation across strategies. The two noncompliance strategies are positively correlated across lenders (Figure B.5), suggesting that noncompliance is a lender-level characteristic that manifests through multiple channels.

Figure B.5: Correlation Between Subsegment Misclassification and Activity Reclassification



Note: Correlation between subsegment misclassification and activity reclassification across lenders.

Table B.1 presents regression estimates that jointly characterize the correlates of each noncompliance strategy. The unit of observation is a lender–market pair. A one-standard-deviation increase in the pre-reform interest rate is associated with a 24.6 percentage-point increase in the subsegment misclassification rate (column 1) and a 3.4 percentage-point increase in the activity reclassification rate (column 2), both statistically significant. A one-standard-deviation increase in the lender's pre-reform market share is associated with statistically significant reductions in both measures. The number of lenders in the market is positively associated with subsegment misclassification.

Table B.1: Correlates of regulatory noncompliance

	Subsegment misclassification (1)	Activity reclassification (2)
Interest rate	0.246***	0.034**
	(0.025)	(0.015)
N lenders	0.052***	-0.037***
	(0.019)	(0.013)
Market share	-0.017*	-0.024***
	(0.010)	(0.007)
Constant	0.418***	0.207***
	(0.028)	(0.016)
Observations	1573	1329
R-squared	0.396	0.033

Notes: This table reports joint correlates of noncompliance strategies. The unit of observation is a lender–market pair. The dependent variable in column (1) is the share of agricultural loans with interest rates above the cap that are classified in non-targeted regulatory subsegments. The dependent variable in column (2) is the change, relative to the pre-reform period, in the share of loans classified under the non-agricultural sector among activity-switched loans. Negative changes are censored at zero, as the targeted sector did not exist prior to the reform. Independent variables are standardized. Interest rates and market shares are measured in the pre-reform period. Standard errors are reported in parentheses and are clustered at the lender and market levels. * $p< 0.10$, ** $p< 0.05$, *** $p< 0.01$

C Targeting of Single-Lender Borrowers

A key distributional finding is that lenders disproportionately direct noncompliance toward borrowers with whom they have an exclusive lending relationship. We define a *single-lender borrower* as a borrower who, among all loans taken over the sample period, has obtained credit from only one lender. Approximately 55 percent of multi-loan borrowers in our sample are single-lender borrowers by this definition.

Table C.1 presents regression estimates of subsegment misclassification on indicators for single-lender status, loan sequence, and their interaction. The coefficient on the interaction term between single-lender status and the indicator for subsequent loans is positive and significant at the one-percent level. The estimate implies that, relative to their baseline probability, subsequent loans to single-lender borrowers are approximately 5 percent (1.9 percentage points) more likely to be misclassified into a non-agricultural subsegment.

Table C.1: Subsegment misclassification and single-lender borrowers

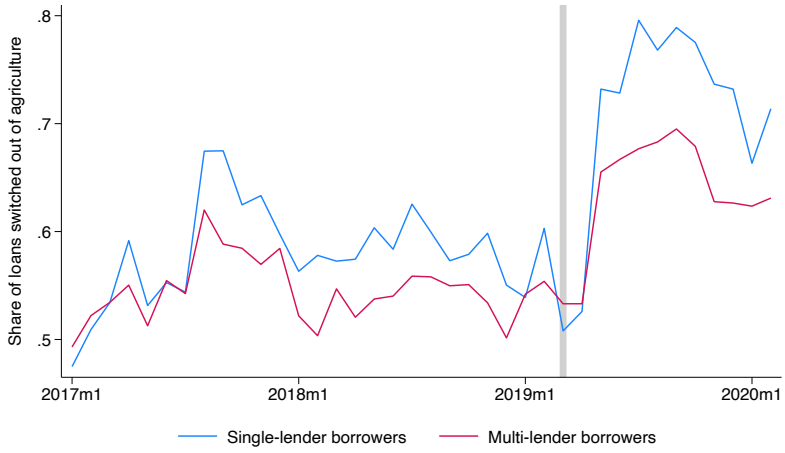
	Subsegment misclassification		
	(1)	(2)	(3)
Single lender	0.000		-0.014**
	(0.003)		(0.006)
Subsequent loan		0.007**	-0.005
		(0.003)	(0.006)
Single lender × Subsequent loan			0.019***
			(0.007)
Constant	0.387***	0.382***	0.391***
	(0.002)	(0.003)	(0.005)
Observations	65,961	65,961	65,961
R-squared	0.543	0.543	0.543

Notes: The dependent variable is an indicator for agricultural loans with interest rates above the cap that are classified in non-targeted regulatory subsegments. Single lender is a indicator for borrowers with multiple loans taken from a single lender. Subsequent loan is an indicator for loans that are not the borrower's first loan. All regressions include lender and month fixed effects. Standard errors are reported in parentheses and are clustered at the borrower level. * p< 0.10, ** p< 0.05, *** p< 0.01

A parallel pattern holds for activity reclassification. Figure C.1 shows that the post-reform

increase in the share of loans switched out of agriculture is substantially larger for single-lender borrowers than for multi-lender borrowers. Table C.2 presents the corresponding difference-in-differences estimates. The post-reform increase in activity reclassification is roughly 33 percent larger for single-lender borrowers than for multi-lender borrowers (16.9 percentage points versus 12.2 percentage points). The interaction between single-lender status and the post-reform indicator is 0.040, significant at the one-percent level (column 3).

Figure C.1: Share of Loans Switched Out of Agriculture by Borrower Type



Note: Share of loans switched out of agriculture for single-lender and multi-lender borrowers, 2017–2020. The gap between the two series widens after the reform, indicating that single-lender borrowers bear a disproportionate share of activity reclassification.

These patterns are consistent with lenders exploiting a captive-borrower margin. Single-lender borrowers have fewer outside options than multi-lender borrowers and therefore face a lower likelihood of switching to a different lender in response to above-cap rates. Heterogeneity in borrower characteristics supports this interpretation: borrowers with multiple lending relationships have on average larger loans, higher income, higher expenditure, and greater wealth, suggesting that access to multiple lenders is associated with stronger bargaining position. Borrowers who switch from agricultural to non-agricultural classifications post-reform receive lower loan amounts and have lower income and wealth than non-switching borrowers.

Table C.2: Activity reclassification and single-lender borrowers

	Switch out of agriculture		
	(1)	(2)	(3)
Post	0.139***		0.122***
	(0.003)		(0.004)
Single lender		0.021***	0.007*
		(0.003)	(0.003)
Single lender × Post			0.040***
			(0.006)
Constant	0.556***	0.591***	0.553***
	(0.002)	(0.002)	(0.002)
Observations	102,696	102,696	102,696
R-squared	0.121	0.105	0.122

Notes: The dependent variable is an indicator for loans classified under the non-agricultural sector among activity-switched loans. Single lender is a indicator for borrowers with multiple loans taken from a single lender. Post is an indicator for the time period when the reform was in place. All regressions include lender fixed effects. Standard errors are reported in parentheses and are clustered at the borrower level. * $p< 0.10$, ** $p< 0.05$, *** $p< 0.01$